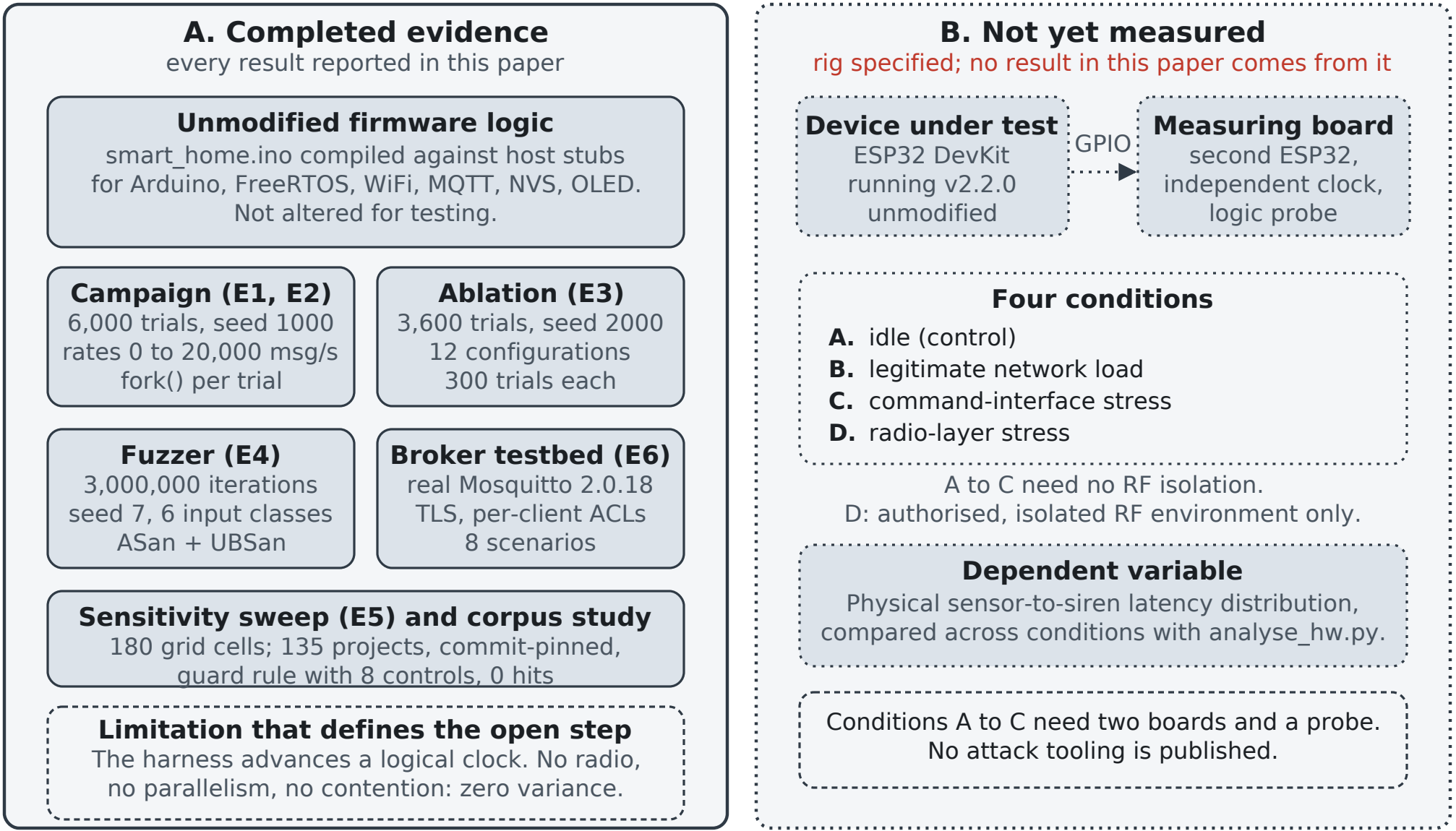
A. Completed evidence
every result reported in this paper
Unmodified firmware logic
smart_home.ino compiled against host stubs
for Arduino, FreeRTOS, WiFi, MQTT, NVS, OLED.
Not altered for testing.
Campaign (E1, E2)
6,000 trials, seed 1000
rates 0 to 20,000 msg/s
fork() per trial
Ablation (E3)
3,600 trials, seed 2000
12 configurations
300 trials each
Fuzzer (E4)
3,000,000 iterations
seed 7, 6 input classes
ASan + UBSan
Broker testbed (E6)
real Mosquitto 2.0.18
TLS, per-client ACLs
8 scenarios
Sensitivity sweep (E5) and corpus study
180 grid cells; 135 projects, commit-pinned,
guard rule with 8 controls, 0 hits
Limitation that defines the open step
The harness advances a logical clock. No radio,
no parallelism, no contention: zero variance.
B. Not yet measured
rig specified; no result in this paper comes from it
Device under test
ESP32 DevKit
running v2.2.0
unmodified
GPIO
Measuring board
second ESP32,
independent clock,
logic probe
Four conditions
A. idle (control)
B. legitimate network load
C. command-interface stress
D. radio-layer stress
A to C need no RF isolation.
D: authorised, isolated RF environment only.
Dependent variable
Physical sensor-to-siren latency distribution,
compared across conditions with analyse_hw.py.
Conditions A to C need two boards and a probe.
No attack tooling is published.